

Question 6.4

Unrefereed candidate disproof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 2510.11311__04 — generated 2026-09-17

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

Summary and provenance

Catalog id	2510.11311__04
Catalog entry	Question 6.4
Source corpus	arxiv
Campaign leg	selected second attempts (<code>attacks_retry</code>)
Source paper / entry	Extending Thomassen's conjecture to directed graphs
Model verdict	disproved (confidence: high)
Model's one-line claim	The anti-directed C4 is not Eulerian-avoidable: Eulerian oriented graphs can have arbitrarily large minimum out-degree while having no anti-directed-C4-free subdigraph of minimum out-degree 2.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	<code>attacks_retry/2510.11311__04/output.md</code>
Independent review	None
Model's caveats	The counterexamples have very large degree spread; this does not classify the other orientations of C4.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

A counterexample for the anti-directed orientation

Let $F = \vec{K}_{2,2}$, with arcs

$$u \rightarrow x, \quad u \rightarrow y, \quad v \rightarrow x, \quad v \rightarrow y.$$

All digraphs below are finite and simple, and copies need not be induced.

Theorem. For every integer $d \geq 2$, there is a strongly connected Eulerian oriented graph D_d with

$$\delta^+(D_d) = d$$

such that no nonempty subdigraph of D_d has both minimum out-degree at least 2 and no copy of F .

Thus the answer to the question is **no**.

The construction combines a small terminal bottleneck with a long sequence of regular, high-girth bipartite layers. Avoiding F bounds the number of surviving vertices immediately before the bottleneck. The high-girth layers then force these numbers to decrease strictly when traced backwards, eventually giving a contradiction. In particular, deleting vertices cannot circumvent the obstruction.

1. A small-target growth lemma

Lemma 1. Let Q be a bipartite graph with parts X, Y and girth greater than $2M$. Suppose that nonempty sets $P \subseteq X$ and $R \subseteq Y$ satisfy

$$|R| \leq M$$

and every vertex of P has at least two neighbors in R . Then

$$|P| \leq |R| - 1.$$

Proof. Any cycle in $Q[P, R]$ would have length at most $2|R| \leq 2M$, because a bipartite cycle uses equally many distinct vertices from each part. Consequently, $Q[P, R]$ is a forest. Therefore

$$2|P| \leq e(Q[P, R]) \leq |P| + |R| - 1,$$

which proves the claim. $\square$

We will use regular bipartite graphs of arbitrarily large prescribed girth. A fully explicit, self-contained construction is given in Section 4.

2. Construction of the Eulerian host

Fix $d \geq 2$, and put

$$M = \binom{d}{2}, \quad L = M + 1.$$

Choose a simple d -regular bipartite graph Q , with n vertices in each part, whose girth is greater than $2M$.

Take disjoint vertex sets

$$B, A_1, A_2, \dots, A_L, \quad |B| = d, \quad |A_i| = n.$$

Define D_d by the following arcs:

1. All arcs from B to A_1 .
2. For each $1 \leq i < L$, a copy of Q between A_i and A_{i+1} , with every edge directed from A_i to A_{i+1} .
3. All arcs from A_L to B .

There are no other arcs. Schematically,

$$B \longrightarrow A_1 \xrightarrow{Q} A_2 \xrightarrow{Q} \dots \xrightarrow{Q} A_L \longrightarrow B.$$

Every vertex of B has in-degree and out-degree n . Every vertex of every A_i has in-degree and out-degree d . Since $n \geq d$,

$$d_{D_d}^-(z) = d_{D_d}^+(z) \quad \text{for all } z, \quad \delta^+(D_d) = d.$$

Also, $L \geq 2$, so this is an oriented graph: there are no antiparallel pairs.

The digraph is strongly connected. Every vertex in an A_i can reach B by following the layers forwards. Conversely, tracing predecessors backwards through the regular layers shows that every vertex in an A_i is reachable from A_1 , and hence from every vertex of B . Thus the construction satisfies even the convention that an Eulerian digraph must be connected.

3. No F -free subdigraph can have minimum out-degree 2

Suppose, for a contradiction, that a nonempty subdigraph $H \subseteq D_d$ is F -free and satisfies

$$\delta^+(H) \geq 2.$$

Set

$$T = V(H) \cap B, \quad S_i = V(H) \cap A_i.$$

All these sets are nonempty. Indeed, every arc goes to the next class in the cyclic sequence

$$B, A_1, \dots, A_L, B.$$

Starting at any vertex of H and repeatedly following an outgoing arc therefore visits every class.

The final layer is small For each $z \in S_L$, choose two distinct out-neighbors of z in H . Both belong to T ; denote their unordered pair by P_z .

If two distinct vertices $z, z' \in S_L$ had $P_z = P_{z'}$, these two sources and their two common heads would form a copy of $\vec{K}_{2,2}$. Hence the pairs P_z are all distinct, and

$$|S_L| \leq \binom{|T|}{2} \leq \binom{d}{2} = M. \quad (3.1)$$

This is the elementary pair-counting obstruction from the supplied lead, now applied at a designated final layer.

Every preceding layer is strictly smaller Suppose $1 \leq i < L$ and $|S_{i+1}| \leq M$. Every vertex of S_i has at least two out-neighbors in S_{i+1} , since these are its only possible out-neighbors in H .

The underlying graph between A_i and A_{i+1} is a copy of Q . Lemma 1 therefore gives

$$|S_i| \leq |S_{i+1}| - 1. \quad (3.2)$$

Starting from (3.1) and applying (3.2) backwards through the $L - 1 = M$ intermediate transitions yields

$$|S_1| \leq |S_L| - (L - 1) \leq M - M = 0,$$

contradicting the nonemptiness of S_1 .

This proves the theorem. Notice that H was not assumed to be spanning or Eulerian.

If an Eulerian-avoidability function d_F existed, choosing

$$d \geq \max\{2, d_F(2)\}$$

and applying its defining property to D_d would contradict the theorem. Therefore $\vec{K}_{2,2}$ is not Eulerian-avoidable. $\square$

4. An explicit construction of the high-girth layers

For completeness, the required regular bipartite graph can be constructed without invoking any external existence theorem.

Lemma 2. For every $d \geq 2$ and $g \geq 2$, one can explicitly construct a finite simple d -regular bipartite graph of girth greater than g .

Proof. Let W consist of the empty word and all words of length at most g over the alphabet $[d]$ in which consecutive letters differ.

For each $j \in [d]$, define a permutation σ_j of W as follows:

- If a word ends in j , delete its final letter.

- Otherwise, if its length is less than g , append j .
- Otherwise, fix it.

Each σ_j is an involution. We use right actions for permutation multiplication.

If

$$j_1 j_2 \cdots j_\ell, \quad 1 \leq \ell \leq g,$$

has consecutive letters distinct, then

$$\emptyset \sigma_{j_1} \sigma_{j_2} \cdots \sigma_{j_\ell} = j_1 j_2 \cdots j_\ell.$$

In particular,

$$\sigma_{j_1} \sigma_{j_2} \cdots \sigma_{j_\ell} \neq 1. \quad (4.1)$$

Let $\Gamma = \text{Sym}(W)$. Form a bipartite graph Q whose two parts are copies Γ_0, Γ_1 of Γ , with edges

$$\pi_0 (\pi \sigma_j)_1 \quad (\pi \in \Gamma, j \in [d]).$$

The generators $\sigma_1, \dots, \sigma_d$ are distinct, so Q is simple and d -regular. Give each displayed edge color j . Each color class is a perfect matching.

Suppose Q had a simple cycle of length $\ell \leq g$. Reading its edge colors in order gives a sequence $j_1, \dots, j_\ell$ with consecutive colors distinct: using the same color twice consecutively would immediately traverse the same matching edge backwards. Since every σ_j is an involution, returning to the starting vertex gives

$$\sigma_{j_1} \sigma_{j_2} \cdots \sigma_{j_\ell} = 1,$$

contrary to (4.1). Hence Q has girth greater than g . $\square$

For the counterexample, take $g = 2M = d(d-1)$. This gives a completely specified family. More explicitly, one may use

$$|W| = 1 + d \sum_{r=0}^{d(d-1)-1} (d-1)^r, \quad n = |W|!,$$

so that

$$|V(D_d)| = d + \left(\binom{d}{2} + 1 \right) n.$$

These sizes are deliberately unoptimized.

5. Scope and a generalization

The same construction proves a broader statement:

Corollary. For all $s, t \geq 2$, the one-directed complete bipartite digraph $\vec{K}_{s,t}$, directed from its s -vertex part to its t -vertex part, is not Eulerian-avoidable.

To see this, fix $d \geq t$, use target minimum out-degree $k = t$, and replace the parameters by

$$M = (s-1) \binom{d}{t}, \quad L = M + 1.$$

In a $\vec{K}_{s,t}$ -free subdigraph, choosing a t -set of out-neighbors for each final-layer vertex assigns any particular t -set to at most $s-1$ sources. Thus the final layer again has at most M vertices. Since $t \geq 2$, Lemma 1 again forces strict growth forwards through every small intermediate layer, giving the same contradiction.

For the original question, only $s = t = 2$ is needed. The crucial distinction from complete cyclic blow-ups is that the intermediate high-girth layers cannot support a small 2-out core. This removes the vertex-deletion escape available in the supplied attempt and gives a counterexample to Eulerian-avoidability itself, not merely to a spanning version.